

Technical Document #492: Software Engineering - Comprehensive Overview

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Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

Design patterns provide reusable solutions to common software design problems. Singleton, factory, and observer patterns are widely used.

Domain-driven design (DDD) models software around business domains. It improves communication between technical and business stakeholders.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Key Takeaways:

- Technical debt represents the cost of choosing easy solutions over better ones.
- Code review examines code changes before merging.
- Refactoring improves code structure without changing behavior.